

Kevin Tran

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EDUCATION

Toronto Metropolitan University

Bachelor of Science in Computer Science (Co-op)

Toronto, Ontario

Sept. 2024 – April 2028

PROJECTS

Rubik's Cube Solver | *Python, HTML, CSS, JavaScript, FastAPI*

- Implemented a layer-by-layer solving system (**CFOP-style**), including **F2L**, **OLL**, and **PLL** case detection, capable of solving any valid scramble across **43 quintillion** possible configurations
- Engineered a comprehensive testing environment using **Pytest** framework, validating solver accuracy across CFOP stages, achieving 100% solve rate over 1,000+ randomized trial scrambles while enforcing a strict move-count efficiency threshold.
- Developed a **REST API** using **FastAPI** to expose solver logic as HTTP endpoints, enabling communication between the Python backend and JavaScript frontend
- Built an interactive web demo featuring real-time **3D cube visualization** and step-by-step solution animation

IMDB Data Insights & Movie Recommender | *Python, Streamlit*

- Developed a **Python-based** platform that recommends movies and TV shows based on user-selected titles, genres, and popularity metrics, processing around **~ 1 million IMDb titles**
- Implemented **sorting and filtering** functionality to list content by rating, number of votes, and release year, enabling dynamic exploration of popular titles
- Processed and analyzed **IMDb datasets** using **pandas** and **NumPy** to handle large-scale data efficiently and generate accurate recommendations

Gengar Simulator | *Java*

- Designed a turn-based simulation game based on the video game "Pokemon"
- Applied **linear algebra** concepts (**points, matrices**) to control **animation** paths and drawing logic
- Built an interactive **GUI** with **JSwing** for animations and to enhance user experience

Infinite Range RC Car (Team Project) | *C++, Linux*

- Collaborated on building a long-range **RC car** using **Raspberry Pi** and embedded components
- Assisted with soldering, wiring, and assembly of motors and power systems
- Worked with **C++** and **Linux** on Raspberry Pi to test and debug vehicle control behavior
- Helped troubleshoot power, connectivity, and stability issues during testing

TECHNICAL SKILLS

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL

Frameworks/Cloud: AWS, Pytest, REST APIs, FastAPI, React, Node.js JSwing

Developer Tools: Linux, Git, VS Code, Excel

Libraries: pandas, NumPy

EXPERIENCE

Coding Instructor

STEM Camp

June 2024 – Sept. 2024

Newmarket, Ontario

- Led hands-on learning sessions focused on coding and robotics, fostering student engagement and technical understanding
- Guided students through practical applications of loops, conditionals, and variables in Python, with practical exercises
- Mentored students in debugging and problem-solving, fostering critical thinking and coding proficiency

Tax Preparer Associate (Co-op)

Liberty Tax

Feb. 2022 – June 2022

Newmarket, Ontario

- Conducted detailed client interviews to gather accurate financial and personal information for tax preparation
- Informed clients of required procedures, documentation, and expected timelines throughout the tax filing process
- Prepared and entered tax return data using **Excel** and company software, ensuring accuracy and compliance with CRA standards